

MUSA NAZIF BOYA

SOFTWARE ENGINEER

CONTACT



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https://github.com/musaBoya

EDUCATION

Bechelors degree 09.2015-12.2020

ERCIYES UNIVERSITY
ENGINEERING FACULTY
Electrical/Electronics Engineering

LANGUAGE

YOKDİL(English) - 75.0

SKILLS

- C
- C++(11/14/17)
- Rust
- CMake
- Git
- Linux
- Bash scripting
- QT
- RESTful APIs
- Docker
- Microservice Architecture
- Agile methodologies
- Nginx
- Ffmpeg
- SQL
- RTOS
- uart-spi-i2c
- Bluetooth BLE
- Wifi protocol
- Embedded GUI
- Logic analyzer
- Oscilloscope

REFERENCES

Metin ÖZBİLEN



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- Metin is Technical Leader of Keklikoğlu plastic company's R&D department.

Kadir ALTINDAG



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- Kadir is my Tech lead from Huawei.

SUMMARY

Almost all of the projects I worked on at Huawei ran on **Linux**, although I also had some projects running on **Windows** and **RTOS**. I mostly used **CMake** in these projects. I used **Git** as the version control system in all the projects I worked on. **Agile Scrum** methods were applied in all our projects, and we generally used **Jira** as the tracking system. In some projects, we also had an internal ticket system that we used. In the Cloud OTT project, I developed with **Docker** and gained experience in **microservice architecture**. I worked with **Nginx** and **ffmpeg** here. Since the Cloud OTT project was a media streaming project, I worked with media streaming protocols such as **HLS**, **Dash**, **MSS**, **SRT**, and **RTMP**. I made improvements with **RESTful APIs**. I used **SQLite** on the database side. I used **QT** and **ImGui** for the interface in my own projects. When I was working on the **embedded** side, I worked with processors such as **ESP32** and **STM32**. I established communication using **UART**, **SPI**, **I2C**, **Wi-Fi**, and **Bluetooth protocols**. I worked with hardware devices such as **motor controllers**, **audio amplifiers**, **valves**, **sensors**, and **pumps**. I performed **debugging analysis** using devices such as **oscilloscopes**, **multimeters**, and **logic analyzers**.

WORK EXPERIENCE

C/C++ SOFTWARE ENGINEER

12.2022-Present

Huawei Turkey R&D Center

WebLMT (11 months) -> Weblmt(Web Based Local Maintenance Terminal) project is a remote access and management software with a web-based interface through which we access the base station. I developed code in 4 different product under Weblmt project. Our products can run on RTOS, Linux and Windows. And we used C and C++ in that project.

Achievements: C, C++, Embedded Linux, patching, Git, Jenkins, Agile Methodologies, Scrum, Software Quality, DTS

Cloud OTT (13 months) -> OTT(over-the-top) project refers to media services that reach the audience directly over the internet. I worked in two projects under ott. In first project, the team responsible for the Microservice related to media flow and broadcast flow in the Ott project. I used Rtmp and Srt protocols with Nginx and used ffmpeg. We also used Docker since it was code running in the cloud. Second project was allowing streamers to easily share and distribute live video streams across different locations and platforms with low latency, scalability, and flexible protocols, including SRT. We used c++17 standards here, it was multi thread and integrated RESTful APIs using HTTP/HTTPS protocols. Built an **internal performance benchmarking tool** that simulates mass client behavior to analyze stream stability and latency. I developed RESTful APIs using HTTP.

Achievements: C, C++, Linux, CMake, Git, Jira, Agile Methodologies, Software Quality, Teamwork

OpenHarmony Project(01.2025 - Present) -> I participated in an open-source project contributing to the compiler and runtime components of the ArkTS language for the OpenHarmony operating system ecosystem. Focused on the development of low-level system components, including the language processing and runtime infrastructure for ArkTS, to optimize compiler performance, enhance runtime execution, and ensure overall language stability. Currently contributing to the Media Library project, focusing on system-level media capabilities and integration.

Achievements: C++, SQLite, ArkTS, Ark Compiler, OpenHarmony, Clang, Linux, Cmake, Git, Unit tests, Jira

EMBEDDED SOFTWARE ENGINEER

07.2021 - 12.2022

Keklikoglu Plastik Insaat San. Ltd. Sti.

Kelikoglu Plastik is a company that produces medical simulation models. The humanoid plastic models we produced here were training models that could be analyzed in many ways, such as heart rhythm, breathing, eye reactions, lung and bowel sounds, and training in applications such as heart massage, artificial respiration, intubation, etc.

I worked on both hardware and software in this company, but my main branch was software. I built baremetal and RTOS systems on the models and made both the pcb designs and software of these systems.

We used Uart, I2C, Spi, Bluetooth ble, Wifi communication protocols in the systems we installed, we drove many I/O elements such as various sensors, motors and valves, and we did things like signal processing, screen driving.

I have done many projects related to heart signals and gained experience in signal processing. I have designed GUI using RA8876 graphics Engine by C coding.